An Introduction to Statistical Learning with Applications in R
Chapter 2 Exercises

We will start by loading the required libraries:

```
library("tidyverse")
library("krulRutils")
library("here")
library("ISLR2")
options(scipen = 999) # Disable scientific notation
```

College dataset

For this dataset, we will print a list of all the "elite universities" where we say that a university is elite if more than half of its students were in the top 10 percentile as highschool students. To do that, first we save the list of elite universities in a table:

```
elite_college_tbl <- read_csv(here("data", "College.csv")) %>%
  rename(college_name = 1) %>%
  mutate(elite = if_else(Top10perc > 50, "Yes", "No")) %>%
  filter(elite == "Yes") %>%
  select(college_name)
```

Now that we have the required list, we simply print this information:

```
elite_college_tbl %>%
  print(n = Inf)
```

```
# A tibble: 78 x 1
   college_name
   <chr>
 1 Agnes Scott College
 2 Amherst College
 3 Barnard College
 4 Birmingham-Southern College
 5 Bowdoin College
 6 Brown University
 7 Bryn Mawr College
 8 Carleton College
 9 Carnegie Mellon University
10 Case Western Reserve University
11 Centre College
12 Claremont McKenna College
13 Colby College
14 College of the Holy Cross
15 College of William and Mary
16 Colorado College
17 Columbia University
18 Dartmouth College
19 Davidson College
20 Drew University
21 Duke University
22 Emory University
23 Fresno Pacific College
24 Furman University
25 Georgetown University
26 Georgia Institute of Technology
27 Grinnell College
28 Grove City College
29 Harvard University
30 Harvey Mudd College
31 Hendrix College
32 Illinois Wesleyan University
33 Johns Hopkins University
34 Macalester College
35 Massachusetts Institute of Technology
36 New York University
37 Niagara University
38 Northwestern University
39 Occidental College
40 Oglethorpe University
```

41 Pepperdine University
42 Polytechnic University
43 Princeton University
44 Rensselaer Polytechnic Institute
45 Rhodes College
46 Sarah Lawrence College
47 Scripps College
48 Smith College
49 Stevens Institute of Technology
50 SUNY at Binghamton
51 SUNY College at Geneseo
52 Transylvania University
53 Trenton State College
54 Trinity University
55 University of California at Berkeley
56 University of California at Irvine
57 University of Chicago
58 University of Florida
59 University of Illinois - Urbana
60 University of Michigan at Ann Arbor
61 University of Minnesota at Morris
62 University of North Carolina at Chapel Hill
63 University of Notre Dame
64 University of Pennsylvania
65 University of Puget Sound
66 University of Rochester
67 University of Virginia
68 Vanderbilt University
69 Vassar College
70 Wake Forest University
71 Washington and Lee University
72 Washington University
73 Wellesley College
74 Wesleyan University
75 Wheaton College IL
76 Williams College
77 Wofford College
78 Yale University

Auto dataset

For this dataset, the "horsepower" variable contains some missing data which makes R take the column as a "character" variable, rather than a "numeric" variable. So we start by cleaning this dataset:

```
auto_tbl <- read_csv(here("data", "Auto.csv")) %>%
  mutate(horsepower = as.numeric(horsepower)) %>%
  drop_na()
```

We nos show a summary of the numeric variables of this dataset:

```
summary_auto_tbl <- summarise_numeric_tidy(auto_tbl)
summary_auto_tbl
```

```
# A tibble: 6 x 9
  statistic   mpg cylinders displacement horsepower weight acceleration  year
  <fct>     <dbl>     <dbl>        <dbl>      <dbl>  <dbl>        <dbl> <dbl>
1 min         9         3           68         46    1613           8    70
2 q1         17         4          105         75    2225.         13.8  73
3 median     22.8       4          151         93.5  2804.         15.5  76
4 mean       23.4       5.47       194.       104.   2978.         15.5  76.0
5 q3         29         8          276.       126    3615.         17.0  79
6 max        46.6       8          455        230    5140          24.8  82
# i 1 more variable: origin <dbl>
```

In particular, to find the mean of the "origin" variable, we can do:

```
summary_auto_tbl %>%
  filter(statistic == "mean") %>%
  select(statistic, origin)
```

```
# A tibble: 1 x 2
  statistic origin
  <fct>      <dbl>
1 mean        1.58
```

Now we want to remove the 10th through 85th observations and repeat the above analysis:

```r
summary_sliced_auto_tbl <- auto_tbl %>%
  slice(-10:-85) %>%
  summarise_numeric_tidy()

summary_sliced_auto_tbl
```

```
# A tibble: 6 x 9
  statistic   mpg cylinders displacement horsepower weight acceleration  year
  <fct>     <dbl>     <dbl>        <dbl>      <dbl>  <dbl>        <dbl> <dbl>
1 min        11        3             68          46   1649          8.5  70
2 q1         18        4            100.         75   2214.        14    75
3 median     24.0      4            146.         90   2792.        15.5  77
4 mean       24.4      5.37         187.        101.  2936.        15.7  77.1
5 q3         30.6      6            250         115   3508         17.3  80
6 max        46.6      8            455         230   4997         24.8  82
# i 1 more variable: origin <dbl>
```

```r
summary_sliced_auto_tbl %>%
  filter(statistic == "mean") %>%
  select(statistic, origin)
```

```
# A tibble: 1 x 2
  statistic origin
  <fct>      <dbl>
1 mean        1.60
```

Finally, we will make a bar plot counting the number of vehicles with a given number of cylinders. We start by creating the corresponding table:

```r
cylinder_frequency_tbl <- auto_tbl %>%
  count(cylinders, name = "frequency") %>%
  mutate(cylinders = factor(cylinders)) %>%
  print(n = Inf)
```

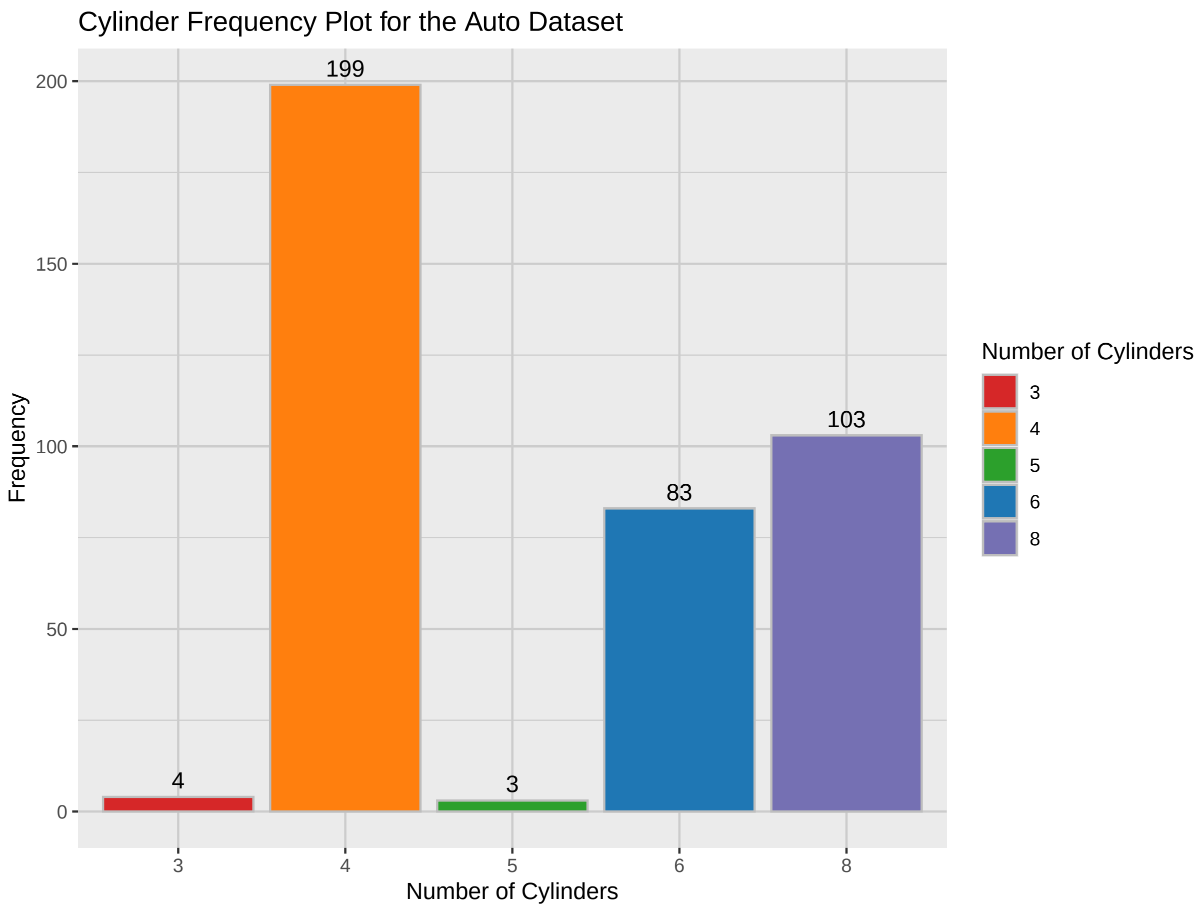
```
# A tibble: 5 x 2
  cylinders frequency
  <fct>         <int>
1 3                 4
2 4               199
3 5                 3
```

```
4 6                 83
5 8                103
```

We now plot this information:

```r
cylinder_colors <- c(
  "#d62728",
  "#ff7f0e",
  "#2ca02c",
  "#1f77b4",
  "#7570b3"
)

cylinder_frequency_plot <- (cylinder_frequency_tbl %>%
  ggplot(
    aes(
      x = cylinders,
      y = frequency,
      fill = cylinders,
      label = frequency
    )
  ) +
  geom_col(color = "gray") +
  geom_text(vjust = -0.5) +
  scale_fill_manual(values = cylinder_colors) +
  theme(
    panel.grid.major = element_line(color = "gray80"),
    panel.grid.minor = element_line(color = "gray80")
  ) +
  labs(
    title = "Cylinder Frequency Plot for the Auto Dataset",
    x = "Number of Cylinders",
    y = "Frequency",
    fill = "Number of Cylinders"
  )) %>%
  print()
```



Boston dataset

This dataset can be obtained directly from the ISLR2 library:

```
data(Boston)
boston_tbl <- tibble(Boston)
```

We start by obtaining a summary of the information contained in this dataset:

```
summary_boston_tbl <- boston_tbl %>%
  summarise_numeric_tidy()

summary_boston_tbl
```

```
# A tibble: 6 x 14
  statistic     crim    zn indus    chas   nox    rm   age   dis   rad   tax
  <fct>        <dbl> <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 min        0.00632   0    0.46 0       0.385  3.56   2.9  1.13  1     187
2 q1         0.0820    0    5.19 0       0.449  5.89  45.0  2.10  4     279
3 median     0.257     0    9.69 0       0.538  6.21  77.5  3.21  5     330
4 mean       3.61     11.4 11.1  0.0692 0.555  6.28  68.6  3.80  9.55  408.
5 q3         3.68     12.5 18.1  0       0.624  6.62  94.1  5.19 24     666
6 max       89.0     100   27.7  1       0.871  8.78 100   12.1  24     711
# i 3 more variables: ptratio <dbl>, lstat <dbl>, medv <dbl>
```

We are now ask for the tracts that have high crime rates, tax rates and pupil-teacher ratios. For this we simply construct the corresponding tables:

```r
q3_boston_tbl <- summary_boston_tbl %>%
  filter(statistic == "q3")

high_crime_rate <- q3_boston_tbl %>%
  pull(crim)

high_tax_rate <- q3_boston_tbl %>%
  pull(tax)

high_student_teacher_ratio <- q3_boston_tbl %>%
  pull(ptratio)

high_crime_boston_tbl <- boston_tbl %>%
  filter(crim > high_crime_rate)

high_tax_boston_tbl <- boston_tbl %>%
  filter(tax > high_tax_rate)

high_student_teacher_tbl <- boston_tbl %>%
  filter(ptratio > high_student_teacher_ratio)
```

Now we can simply print these tables:

```r
high_crime_boston_tbl %>%
  print(n = Inf)
```

```
# A tibble: 127 x 13
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	lstat
	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
1	4.10	0	19.6	0	0.871	5.47	100	1.41	5	403	14.7	26.4
2	8.98	0	18.1	1	0.77	6.21	97.4	2.12	24	666	20.2	17.6
3	3.85	0	18.1	1	0.77	6.40	91	2.51	24	666	20.2	13.3
4	5.20	0	18.1	1	0.77	6.13	83.4	2.72	24	666	20.2	11.5
5	4.26	0	18.1	0	0.77	6.11	81.3	2.51	24	666	20.2	12.7
6	4.54	0	18.1	0	0.77	6.40	88	2.52	24	666	20.2	7.79
7	3.84	0	18.1	0	0.77	6.25	91.1	2.30	24	666	20.2	14.2
8	3.68	0	18.1	0	0.77	5.36	96.2	2.10	24	666	20.2	10.2
9	4.22	0	18.1	1	0.77	5.80	89	1.90	24	666	20.2	14.6
10	4.56	0	18.1	0	0.718	3.56	87.9	1.61	24	666	20.2	7.12
11	3.70	0	18.1	0	0.718	4.96	91.4	1.75	24	666	20.2	14
12	13.5	0	18.1	0	0.631	3.86	100	1.51	24	666	20.2	13.3
13	4.90	0	18.1	0	0.631	4.97	100	1.33	24	666	20.2	3.26
14	5.67	0	18.1	1	0.631	6.68	96.8	1.36	24	666	20.2	3.73
15	6.54	0	18.1	1	0.631	7.02	97.5	1.20	24	666	20.2	2.96
16	9.23	0	18.1	0	0.631	6.22	100	1.17	24	666	20.2	9.53
17	8.27	0	18.1	1	0.668	5.88	89.6	1.13	24	666	20.2	8.88
18	11.1	0	18.1	0	0.668	4.91	100	1.17	24	666	20.2	34.8
19	18.5	0	18.1	0	0.668	4.14	100	1.14	24	666	20.2	38.0
20	19.6	0	18.1	0	0.671	7.31	97.9	1.32	24	666	20.2	13.4
21	15.3	0	18.1	0	0.671	6.65	93.3	1.34	24	666	20.2	23.2
22	9.82	0	18.1	0	0.671	6.79	98.8	1.36	24	666	20.2	21.2
23	23.6	0	18.1	0	0.671	6.38	96.2	1.39	24	666	20.2	23.7
24	17.9	0	18.1	0	0.671	6.22	100	1.39	24	666	20.2	21.8
25	89.0	0	18.1	0	0.671	6.97	91.9	1.42	24	666	20.2	17.2
26	15.9	0	18.1	0	0.671	6.54	99.1	1.52	24	666	20.2	21.1
27	9.19	0	18.1	0	0.7	5.54	100	1.58	24	666	20.2	23.6
28	7.99	0	18.1	0	0.7	5.52	100	1.53	24	666	20.2	24.6
29	20.1	0	18.1	0	0.7	4.37	91.2	1.44	24	666	20.2	30.6
30	16.8	0	18.1	0	0.7	5.28	98.1	1.43	24	666	20.2	30.8
31	24.4	0	18.1	0	0.7	4.65	100	1.47	24	666	20.2	28.3
32	22.6	0	18.1	0	0.7	5	89.5	1.52	24	666	20.2	32.0
33	14.3	0	18.1	0	0.7	4.88	100	1.59	24	666	20.2	30.6
34	8.15	0	18.1	0	0.7	5.39	98.9	1.73	24	666	20.2	20.8
35	6.96	0	18.1	0	0.7	5.71	97	1.93	24	666	20.2	17.1
36	5.29	0	18.1	0	0.7	6.05	82.5	2.17	24	666	20.2	18.8
37	11.6	0	18.1	0	0.7	5.04	97	1.77	24	666	20.2	25.7
38	8.64	0	18.1	0	0.693	6.19	92.6	1.79	24	666	20.2	15.2
39	13.4	0	18.1	0	0.693	5.89	94.7	1.78	24	666	20.2	16.4
40	8.72	0	18.1	0	0.693	6.47	98.8	1.73	24	666	20.2	17.1
41	5.87	0	18.1	0	0.693	6.40	96	1.68	24	666	20.2	19.4

42	7.67	0	18.1	0	0.693	5.75	98.9	1.63	24	666	20.2	19.9
43	38.4	0	18.1	0	0.693	5.45	100	1.49	24	666	20.2	30.6
44	9.92	0	18.1	0	0.693	5.85	77.8	1.50	24	666	20.2	30.0
45	25.0	0	18.1	0	0.693	5.99	100	1.59	24	666	20.2	26.8
46	14.2	0	18.1	0	0.693	6.34	100	1.57	24	666	20.2	20.3
47	9.60	0	18.1	0	0.693	6.40	100	1.64	24	666	20.2	20.3
48	24.8	0	18.1	0	0.693	5.35	96	1.70	24	666	20.2	19.8
49	41.5	0	18.1	0	0.693	5.53	85.4	1.61	24	666	20.2	27.4
50	67.9	0	18.1	0	0.693	5.68	100	1.43	24	666	20.2	23.0
51	20.7	0	18.1	0	0.659	4.14	100	1.18	24	666	20.2	23.3
52	12.0	0	18.1	0	0.659	5.61	100	1.29	24	666	20.2	12.1
53	7.40	0	18.1	0	0.597	5.62	97.9	1.45	24	666	20.2	26.4
54	14.4	0	18.1	0	0.597	6.85	100	1.47	24	666	20.2	19.8
55	51.1	0	18.1	0	0.597	5.76	100	1.41	24	666	20.2	10.1
56	14.1	0	18.1	0	0.597	6.66	100	1.53	24	666	20.2	21.2
57	18.8	0	18.1	0	0.597	4.63	100	1.55	24	666	20.2	34.4
58	28.7	0	18.1	0	0.597	5.16	100	1.59	24	666	20.2	20.1
59	45.7	0	18.1	0	0.693	4.52	100	1.66	24	666	20.2	37.0
60	18.1	0	18.1	0	0.679	6.43	100	1.83	24	666	20.2	29.0
61	10.8	0	18.1	0	0.679	6.78	90.8	1.82	24	666	20.2	25.8
62	25.9	0	18.1	0	0.679	5.30	89.1	1.65	24	666	20.2	26.6
63	73.5	0	18.1	0	0.679	5.96	100	1.80	24	666	20.2	20.6
64	11.8	0	18.1	0	0.718	6.82	76.5	1.79	24	666	20.2	22.7
65	11.1	0	18.1	0	0.718	6.41	100	1.86	24	666	20.2	15.0
66	7.02	0	18.1	0	0.718	6.01	95.3	1.87	24	666	20.2	15.7
67	12.0	0	18.1	0	0.614	5.65	87.6	1.95	24	666	20.2	14.1
68	7.05	0	18.1	0	0.614	6.10	85.1	2.02	24	666	20.2	23.3
69	8.79	0	18.1	0	0.584	5.56	70.6	2.06	24	666	20.2	17.2
70	15.9	0	18.1	0	0.679	5.90	95.4	1.91	24	666	20.2	24.4
71	12.2	0	18.1	0	0.584	5.84	59.7	2.00	24	666	20.2	15.7
72	37.7	0	18.1	0	0.679	6.20	78.7	1.86	24	666	20.2	14.5
73	7.37	0	18.1	0	0.679	6.19	78.1	1.94	24	666	20.2	21.5
74	9.34	0	18.1	0	0.679	6.38	95.6	1.97	24	666	20.2	24.1
75	8.49	0	18.1	0	0.584	6.35	86.1	2.05	24	666	20.2	17.6
76	10.1	0	18.1	0	0.584	6.83	94.3	2.09	24	666	20.2	19.7
77	6.44	0	18.1	0	0.584	6.42	74.8	2.20	24	666	20.2	12.0
78	5.58	0	18.1	0	0.713	6.44	87.9	2.32	24	666	20.2	16.2
79	13.9	0	18.1	0	0.713	6.21	95	2.22	24	666	20.2	15.2
80	11.2	0	18.1	0	0.74	6.63	94.6	2.12	24	666	20.2	23.3
81	14.4	0	18.1	0	0.74	6.46	93.3	2.00	24	666	20.2	18.0
82	15.2	0	18.1	0	0.74	6.15	100	1.91	24	666	20.2	26.4
83	13.7	0	18.1	0	0.74	5.94	87.9	1.82	24	666	20.2	34.0
84	9.39	0	18.1	0	0.74	5.63	93.9	1.82	24	666	20.2	22.9

85	22.1	0	18.1	0	0.74	5.82	92.4	1.87	24	666	20.2	22.1
86	9.72	0	18.1	0	0.74	6.41	97.2	2.07	24	666	20.2	19.5
87	5.67	0	18.1	0	0.74	6.22	100	2.00	24	666	20.2	16.6
88	9.97	0	18.1	0	0.74	6.48	100	1.98	24	666	20.2	18.8
89	12.8	0	18.1	0	0.74	5.85	96.6	1.90	24	666	20.2	23.8
90	10.7	0	18.1	0	0.74	6.46	94.8	1.99	24	666	20.2	24.0
91	6.29	0	18.1	0	0.74	6.34	96.4	2.07	24	666	20.2	17.8
92	9.92	0	18.1	0	0.74	6.25	96.6	2.20	24	666	20.2	16.4
93	9.33	0	18.1	0	0.713	6.18	98.7	2.26	24	666	20.2	18.1
94	7.53	0	18.1	0	0.713	6.42	98.3	2.18	24	666	20.2	19.3
95	6.72	0	18.1	0	0.713	6.75	92.6	2.32	24	666	20.2	17.4
96	5.44	0	18.1	0	0.713	6.66	98.2	2.36	24	666	20.2	17.7
97	5.09	0	18.1	0	0.713	6.30	91.8	2.37	24	666	20.2	17.3
98	8.25	0	18.1	0	0.713	7.39	99.3	2.45	24	666	20.2	16.7
99	9.51	0	18.1	0	0.713	6.73	94.1	2.50	24	666	20.2	18.7
100	4.75	0	18.1	0	0.713	6.52	86.5	2.44	24	666	20.2	18.1
101	4.67	0	18.1	0	0.713	5.98	87.9	2.58	24	666	20.2	19.0
102	8.20	0	18.1	0	0.713	5.94	80.3	2.78	24	666	20.2	16.9
103	7.75	0	18.1	0	0.713	6.30	83.7	2.78	24	666	20.2	16.2
104	6.80	0	18.1	0	0.713	6.08	84.4	2.72	24	666	20.2	14.7
105	4.81	0	18.1	0	0.713	6.70	90	2.60	24	666	20.2	16.4
106	3.69	0	18.1	0	0.713	6.38	88.4	2.57	24	666	20.2	14.6
107	6.65	0	18.1	0	0.713	6.32	83	2.73	24	666	20.2	14.0
108	5.82	0	18.1	0	0.713	6.51	89.9	2.80	24	666	20.2	10.3
109	7.84	0	18.1	0	0.655	6.21	65.4	2.96	24	666	20.2	13.2
110	3.77	0	18.1	0	0.655	5.95	84.7	2.87	24	666	20.2	17.2
111	4.42	0	18.1	0	0.584	6.00	94.5	2.54	24	666	20.2	21.3
112	15.6	0	18.1	0	0.58	5.93	71	2.91	24	666	20.2	18.1
113	13.1	0	18.1	0	0.58	5.71	56.7	2.82	24	666	20.2	14.8
114	4.35	0	18.1	0	0.58	6.17	84	3.03	24	666	20.2	16.3
115	4.04	0	18.1	0	0.532	6.23	90.7	3.10	24	666	20.2	12.9
116	4.65	0	18.1	0	0.614	6.98	67.6	2.53	24	666	20.2	11.7
117	8.06	0	18.1	0	0.584	5.43	95.4	2.43	24	666	20.2	18.1
118	6.39	0	18.1	0	0.584	6.16	97.4	2.21	24	666	20.2	24.1
119	4.87	0	18.1	0	0.614	6.48	93.6	2.31	24	666	20.2	18.7
120	15.0	0	18.1	0	0.614	5.30	97.3	2.10	24	666	20.2	24.9
121	10.2	0	18.1	0	0.614	6.18	96.7	2.17	24	666	20.2	18.0
122	14.3	0	18.1	0	0.614	6.23	88	1.95	24	666	20.2	13.1
123	5.82	0	18.1	0	0.532	6.24	64.7	3.42	24	666	20.2	10.7
124	5.71	0	18.1	0	0.532	6.75	74.9	3.33	24	666	20.2	7.74
125	5.73	0	18.1	0	0.532	7.06	77	3.41	24	666	20.2	7.01
126	5.69	0	18.1	0	0.583	6.11	79.8	3.55	24	666	20.2	15.0
127	4.84	0	18.1	0	0.583	5.90	53.2	3.15	24	666	20.2	11.4

```
# i 1 more variable: medv <dbl>
```

```r
high_tax_boston_tbl %>%
  print(n = Inf)
```

```
# A tibble: 5 x 13
   crim    zn indus  chas   nox    rm   age   dis   rad   tax ptratio lstat
  <dbl> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <int> <dbl>   <dbl> <dbl>
1 0.151     0  27.7     0 0.609  5.45  92.7  1.82     4   711    20.1  18.1
2 0.183     0  27.7     0 0.609  5.41  98.3  1.76     4   711    20.1  24.0
3 0.207     0  27.7     0 0.609  5.09  98    1.82     4   711    20.1  29.7
4 0.106     0  27.7     0 0.609  5.98  98.8  1.87     4   711    20.1  18.1
5 0.111     0  27.7     0 0.609  5.98  83.5  2.11     4   711    20.1  13.4
# i 1 more variable: medv <dbl>
```

```r
high_student_teacher_tbl %>%
  print(n = Inf)
```

```
# A tibble: 56 x 13
     crim    zn indus  chas   nox    rm   age   dis   rad   tax ptratio lstat
    <dbl> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <int> <dbl>   <dbl> <dbl>
 1 0.630      0  8.14     0 0.538  5.95  61.8  4.71     4   307      21  8.26
 2 0.638      0  8.14     0 0.538  6.10  84.5  4.46     4   307      21 10.3
 3 0.627      0  8.14     0 0.538  5.83  56.5  4.50     4   307      21  8.47
 4 1.05       0  8.14     0 0.538  5.94  29.3  4.50     4   307      21  6.58
 5 0.784      0  8.14     0 0.538  5.99  81.7  4.26     4   307      21 14.7
 6 0.803      0  8.14     0 0.538  5.46  36.6  3.80     4   307      21 11.7
 7 0.726      0  8.14     0 0.538  5.73  69.5  3.80     4   307      21 11.3
 8 1.25       0  8.14     0 0.538  5.57  98.1  3.80     4   307      21 21.0
 9 0.852      0  8.14     0 0.538  5.96  89.2  4.01     4   307      21 13.8
10 1.23       0  8.14     0 0.538  6.14  91.7  3.98     4   307      21 18.7
11 0.988      0  8.14     0 0.538  5.81 100    4.10     4   307      21 19.9
12 0.750      0  8.14     0 0.538  5.92  94.1  4.40     4   307      21 16.3
13 0.841      0  8.14     0 0.538  5.60  85.7  4.45     4   307      21 16.5
14 0.672      0  8.14     0 0.538  5.81  90.3  4.68     4   307      21 14.8
15 0.956      0  8.14     0 0.538  6.05  88.8  4.45     4   307      21 17.3
16 0.773      0  8.14     0 0.538  6.50  94.4  4.45     4   307      21 12.8
17 1.00       0  8.14     0 0.538  6.67  87.3  4.24     4   307      21 12.0
18 1.13       0  8.14     0 0.538  5.71  94.1  4.23     4   307      21 22.6
19 1.35       0  8.14     0 0.538  6.07 100    4.18     4   307      21 13.0
20 1.39       0  8.14     0 0.538  5.95  82    3.99     4   307      21 27.7
```

```
21 1.15       0  8.14     0 0.538  5.70  95    3.79     4   307    21   18.4
22 1.61       0  8.14     0 0.538  6.10  96.9  3.76     4   307    21   20.3
23 0.0136    75  4        0 0.41   5.89  47.6  7.32     3   469    21.1 14.8
24 0.149      0  8.56     0 0.52   6.73  79.9  2.78     5   384    20.9  9.42
25 0.114      0  8.56     0 0.52   6.78  71.3  2.86     5   384    20.9  7.67
26 0.229      0  8.56     0 0.52   6.40  85.4  2.71     5   384    20.9 10.6
27 0.212      0  8.56     0 0.52   6.14  87.4  2.71     5   384    20.9 13.4
28 0.140      0  8.56     0 0.52   6.17  90    2.42     5   384    20.9 12.3
29 0.133      0  8.56     0 0.52   5.85  96.7  2.11     5   384    20.9 16.5
30 0.171      0  8.56     0 0.52   5.84  91.9  2.21     5   384    20.9 18.7
31 0.131      0  8.56     0 0.52   6.13  85.2  2.12     5   384    20.9 14.1
32 0.128      0  8.56     0 0.52   6.47  97.1  2.43     5   384    20.9 12.3
33 0.264      0  8.56     0 0.52   6.23  91.2  2.55     5   384    20.9 15.6
34 0.108      0  8.56     0 0.52   6.20  54.4  2.78     5   384    20.9 13
35 0.259      0 21.9      0 0.624  5.69  96    1.79     4   437    21.2 17.2
36 0.325      0 21.9      0 0.624  6.43  98.8  1.81     4   437    21.2 15.4
37 0.881      0 21.9      0 0.624  5.64  94.7  1.98     4   437    21.2 18.3
38 0.340      0 21.9      0 0.624  6.46  98.9  2.12     4   437    21.2 12.6
39 1.19       0 21.9      0 0.624  6.33  97.7  2.27     4   437    21.2 12.3
40 0.590      0 21.9      0 0.624  6.37  97.9  2.33     4   437    21.2 11.1
41 0.330      0 21.9      0 0.624  5.82  95.4  2.47     4   437    21.2 15.0
42 0.976      0 21.9      0 0.624  5.76  98.4  2.35     4   437    21.2 17.3
43 0.558      0 21.9      0 0.624  6.34  98.2  2.11     4   437    21.2 17.0
44 0.323      0 21.9      0 0.624  5.94  93.5  1.97     4   437    21.2 16.9
45 0.352      0 21.9      0 0.624  6.45  98.4  1.85     4   437    21.2 14.6
46 0.250      0 21.9      0 0.624  5.86  98.2  1.67     4   437    21.2 21.3
47 0.545      0 21.9      0 0.624  6.15  97.9  1.67     4   437    21.2 18.5
48 0.291      0 21.9      0 0.624  6.17  93.6  1.61     4   437    21.2 24.2
49 1.63       0 21.9      0 0.624  5.02 100    1.44     4   437    21.2 34.4
50 0.0430    80  1.91     0 0.413  5.66  21.9 10.6      4   334    22    8.05
51 0.107     80  1.91     0 0.413  5.94  19.5 10.6      4   334    22    5.57
52 0.0626     0 11.9      0 0.573  6.59  69.1  2.48     1   273    21    9.67
53 0.0453     0 11.9      0 0.573  6.12  76.7  2.29     1   273    21    9.08
54 0.0608     0 11.9      0 0.573  6.98  91    2.17     1   273    21    5.64
55 0.110      0 11.9      0 0.573  6.79  89.3  2.39     1   273    21    6.48
56 0.0474     0 11.9      0 0.573  6.03  80.8  2.50     1   273    21    7.88
# i 1 more variable: medv <dbl>
```

We now want to compute how many tracts bound the Charles river, for this we simply compute:

```r
boston_tbl %>%
  filter(chas == 1) %>%
  nrow()
```

```
[1] 35
```

To find the median student-teacher ratio among these tracts, we simply use the data that we have already stored in our summary table:

```r
summary_boston_tbl %>%
  filter(statistic == "median") %>%
  select(statistic, ptratio)
```

```
# A tibble: 1 x 2
  statistic ptratio
  <fct>       <dbl>
1 median       19.0
```

We are now asked for the tracts that average more than 7 rooms per dwelling, so we simply count:

```r
boston_tbl %>%
  filter(rm > 7) %>%
  nrow()
```

```
[1] 64
```

Finally, we want to make a comment on the tracts that average more than 8 rooms per dwelling. For that we will simply construct the corresponding table and print it:

```r
high_room_boston_tbl <- boston_tbl %>%
  filter(rm > 8) %>%
  print(n = Inf)
```

```
# A tibble: 13 x 13
     crim    zn indus  chas   nox    rm   age   dis   rad   tax ptratio lstat
    <dbl> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <int> <dbl>   <dbl> <dbl>
 1 0.121      0  2.89     0 0.445  8.07  76    3.50     2   276    18    4.21
 2 1.52       0 19.6      1 0.605  8.38  93.9  2.16     5   403    14.7  3.32
```

```
 3 0.0201    95   2.68      0 0.416  8.03  31.9  5.12     4   224     14.7  2.88
 4 0.315      0   6.2       0 0.504  8.27  78.3  2.89     8   307     17.4  4.14
 5 0.527      0   6.2       0 0.504  8.72  83    2.89     8   307     17.4  4.63
 6 0.382      0   6.2       0 0.504  8.04  86.5  3.22     8   307     17.4  3.13
 7 0.575      0   6.2       0 0.507  8.34  73.3  3.84     8   307     17.4  2.47
 8 0.331      0   6.2       0 0.507  8.25  70.4  3.65     8   307     17.4  3.95
 9 0.369     22   5.86      0 0.431  8.26   8.4  8.91     7   330     19.1  3.54
10 0.612     20   3.97      0 0.647  8.70  86.9  1.80     5   264     13    5.12
11 0.520     20   3.97      0 0.647  8.40  91.5  2.29     5   264     13    5.91
12 0.578     20   3.97      0 0.575  8.30  67    2.42     5   264     13    7.44
13 3.47       0  18.1       1 0.718  8.78  82.9  1.90    24   666     20.2  5.29
# i 1 more variable: medv <dbl>
```

Finally, to find the tracts whose houses have the lowest mean value, we construct the corresponding table and print it:

```
lowest_value_boston_tbl <- boston_tbl %>%
  slice_min(medv, with_ties = TRUE) %>%
  print(n = Inf)
```

```
# A tibble: 2 x 13
   crim    zn indus  chas   nox    rm   age   dis   rad   tax ptratio lstat
  <dbl> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <int> <dbl>   <dbl> <dbl>
1  38.4     0  18.1     0 0.693  5.45   100  1.49    24   666    20.2  30.6
2  67.9     0  18.1     0 0.693  5.68   100  1.43    24   666    20.2  23.0
# i 1 more variable: medv <dbl>
```